A Neuroscience-Grounded Framework for Cognitive Self-Governance: Source Monitoring, Metacognitive Architecture, and Systematic Skill Training

Preprint — Submitted for Expert Review

[Author contact withheld for blind review]

Correspondence: *[withheld for review]*

March 2026

Abstract

The capacity to function as an autonomous cognitive agent — to evaluate information without being subconsciously redirected by its presentation, emotional loading, or repeated exposure — is increasingly recognised as a trainable set of skills rather than a fixed personality characteristic. This preprint presents the *Mental Mastery* framework: a three-layer, 41-skill taxonomy for developing cognitive self-governance, reviewed against 33 peer-reviewed papers spanning mindfulness neuroscience, source monitoring theory, metacognition research, neuroplasticity, and sensory prosthetics. The framework progresses through Foundational Literacy (somatic intelligence, emotion granularity, attention), Metacognitive Monitoring (cognitive process monitoring, emotional self-monitoring, self-boundary integrity, meta-attention), and Intentional Plasticity (state modulation, pattern work), with four cross-layer integration skills. The three-layer sequence is argued to be neurologically non-arbitrary: Layer 2 metacognitive monitoring requires Layer 1 interoceptive signal quality; Layer 3 pattern modification requires Layer 2 monitoring reliability. The Self-Boundary Integrity domain is explicitly grounded in source monitoring theory (Johnson et al., 1993; Mitchell and Johnson, 2009; Simons et al., 2008), identifying five documented failure modes — source amnesia, misattributed familiarity, cryptomnesia, fiction-as-fact incorporation, and criteria relaxation under load — as the principal vulnerabilities the framework trains against. A critical structural constraint is identified: Raio et al. (2013) demonstrated experimentally that prefrontal-dependent regulation strategies trained under calm conditions produce

zero transfer to conditions of acute physiological arousal, with implications for how skills must be assessed and where training must occur. Domain-specificity of metacognition (Morales et al., 2018; Fleming et al., 2014) is incorporated as a second critical constraint, requiring that each monitoring domain receive independent training and assessment rather than assuming transfer from adjacent skills. The framework distinguishes claims that are well-supported by convergent evidence, claims that are mechanistically plausible but not yet integrated-system tested, and claims that exceed current evidence. We present this framework as a structured hypothesis requiring empirical validation and invite expert review of the mechanistic claims, assessment design, and safety provisions.

Keywords: metacognition; source monitoring; neuroplasticity; cognitive self-governance; mindfulness-based training; interoception; prefrontal cortex; probabilistic monitoring architecture; stress-inoculation; domain-specificity

Contents

1 Introduction **5**

2 Framework Architecture **6**

2.1 Taxonomic Status and Epistemic Limitations 7

3 Neuroscientific Foundations **8**

3.1 Neuroplasticity 8

3.2 Attention Regulation Networks 9

3.3 Interoception 9

3.4 Emotional Processing and Granularity 10

3.5 Default Mode Network and Self-Referential Processing 10

3.6 Metacognition 11

3.7 Neuromodulation and the Conditions for Plasticity 11

3.8 Source Monitoring: The Cognitive Mechanism of Self-Boundary Integrity . . . 12

3.9 Two-Tier Metacognitive Architecture and Domain-Specificity 14

4 The Three-Layer Sequence: Neurological Justification **15**

4.1 Layer 1 to Layer 2 Dependency 15

4.2 Layer 2 to Layer 3 Dependency 15

4.3 The L2-to-L3 Gate as a Safety Requirement 16

5 Probabilistic Monitoring Architecture: Channels, Mechanisms, and Known Failure Modes **16**

5.1 Channels of Subconscious Influence 16

5.2 Detection and Resistance by Channel 17

5.3 The Stress-Transfer Problem 17

5.4 Additional Structural Limits 18

6 Critical Design Constraints **19**

6.1 Domain-Specificity of Metacognition 19

6.2 Metacognitive Unavailability Under Working Memory Load 19
6.3 Behavioural Relevance as a Plasticity Gate 19
6.4 The BCT Inversion 20

7 Training Design Principles **20**

8 Assessment Framework **21**

9 Individual Differences and Plasticity Variation **23**

10 Honest Boundaries **23**
10.1 Supported Claims 23
10.2 Plausible But Not Strongly Confirmed 24
10.3 Claims Exceeding Current Evidence 25
10.4 Publication Bias 25

11 Convergence Summary **26**

12 Conclusion **26**

Request for Expert Review **26**

1. Introduction

A practitioner who consistently detects shifts in their own cognitive state — arising from framing effects, social influence, narrative exposure, or emotional contagion — before those shifts complete demonstrates measurably enhanced monitoring reliability under load compared with untrained individuals. The question of whether this monitoring capacity is systematically trainable, and if so how, is both a scientific and a practical one. Exposure conditions that activate documented source monitoring failure modes — repeated framing, narrative embedding, arousal-mediated criteria relaxation, and fluency-based familiarity effects — are well-characterised mechanisms through which information environments exploit known gaps in human source attribution (Johnson et al., 1993). The capacity to resist these influences depends on cognitive skills that, while trainable in principle, have no systematic development framework grounded in neuroscience.

Existing mindfulness-based interventions (MBIs), though well-studied, are developed primarily as therapeutic tools for stress reduction, depression relapse prevention, and attention regulation (Gkintoni et al., 2025; Calderone et al., 2024). They were not designed to produce cognitive self-governance in the specific sense developed here: the real-time detection and attribution of the origins of one's own thoughts, beliefs, and emotional states, maintained under conditions of social pressure, emotional activation, and physiological arousal. The result is that while MBI evidence is relevant and is drawn upon extensively in this framework, it cannot simply be adopted wholesale. Key findings — including the stress-transfer problem (Raio et al., 2013), the domain-specificity of metacognitive accuracy (Morales et al., 2018; Fleming et al., 2014), and the behavioural relevance gate for neuroplasticity (Fallon et al., 2009; Beyeler et al., 2017) — substantially constrain what MBI training produces and what additional training is required.

This preprint presents the Mental Mastery framework: a 41-skill taxonomy organised into three functional layers and nine domains, reviewed against 33 peer-reviewed papers. Section 2 describes the framework's architecture. Section 3 provides the neuroscientific foundation for each core mechanism. Section 4 justifies the three-layer sequence as neurologically non-arbitrary. Section 5 grounds the probabilistic monitoring architecture in source monitoring theory and documents its structural limits. Section 6 describes critical design constraints that emerged from the evidence review. Sections 7–8 describe training and assessment design. Section 9 addresses individual differences. Section 10 states honest boundaries. Section 11 summarises convergence across the 33 papers. Section 12 concludes, and an unnumbered section presents a formal request for expert review.

2. Framework Architecture

The framework is organised into three layers and nine domains, with four cross-layer skills (Table 1).

Table 1: Framework Architecture: Three Layers, Nine Domains, 41 Skills

Layer	Domain	N	Representative Skills
L1: Foundational	Somatic Intelligence	4	Sensation Recognition, Interoceptive Accuracy, Physiological Baseline Mapping, Arousal Detection
L1: Foundational	Emotion Granularity	4	Primary Identification, Somatic-Emotion Mapping, Vocabulary Expansion, Gradient Detection
L1: Foundational	Attention	5	Sustaining, Directing, Recovery, Distraction Resistance, Aperture Control
L2: Metacognitive	Cognitive Process Monitoring	6	Thought Observation, Cognitive State Labelling, Process Awareness, Load Sensing, Assumption Detection, Narrative Monitoring
L2: Metacognitive	Emotional Self-Monitoring	3	Source Attribution, Contagion Awareness, Affect-Cognition Coupling Detection
L2: Metacognitive	Self-Boundary Integrity	4	Self-Other Distinction, Incongruence Detection, Value Anchoring, Identity Stability Monitoring
L2: Metacognitive	Meta-Attention	2	Meta-Attention, Dual Presence
L3: Intentional	State Modulation	4	Voluntary Relaxation, State Shifting, Arousal Calibration, Emotional Response Modification
L3: Intentional	Pattern Work	5	Pattern Recognition, Pattern Interruption, Deliberate Reframing, Topology Mapping, Directed Neuroplastic Practice
Cross-Layer	Integration & Resilience	4	Skill-Under-Load, Recovery Speed, Calibration Honesty, Practice Quality Monitoring
Total		**41**	

The design principles underlying this architecture are:

1. Skills are trainable mechanisms with observable behavioural correlates, not personality traits or fixed capacities.

2. The layer sequence is neurologically motivated: each upper layer requires the lower layer's outputs as inputs. Layers describe *functional dependencies*, not strictly sequential phases.

3. Two independent measurement tracks run throughout: Track A (subjective and behavioural self-report) and Track B (physiological and performance measures). These are explicitly *not* treated as equivalent, following the finding that neural and behavioural improvements are uncorrelated within the same measurement window (Yue et al., 2023).

4. Domain-specificity of metacognition (Morales et al., 2018; Fleming et al., 2014) means each of the four Layer 2 monitoring domains requires independent training and independent assessment. High skill in one domain cannot be assumed to substitute for another.

5. Entry is not fixed. Phase 0 assessment (Toronto Alexithymia Scale (Schandry, 1981), Heartbeat Detection Task, FFMQ Observing Subscale) determines the individual's starting position in Layer 1 before any training begins.

2.1 Taxonomic Status and Epistemic Limitations

The 41-skill count and nine-domain structure require explicit epistemic qualification. The taxonomy was not derived through factor analysis, latent variable modelling, or psychometric validation of a candidate item pool. No confirmatory factor analysis has established that the nine domains correspond to empirically separable factors, and no item-response theory modelling has validated the skill descriptions as a measurement instrument. The following three claims should therefore be understood as the framework's current standing:

1. **The taxonomy is provisional.** Skill boundaries and layer assignments were determined by convergent reading of neuroscientific mechanism and clinical observation, not by data-driven reduction. Skills may merge, split, or be reordered as empirical work proceeds. The 41-skill count is not a stable target; it is a working hypothesis about the granularity at which mechanism-level distinctions become training-relevant.

2. **The skill count is heuristic.** 41 reflects the level of analysis at which each skill has a distinguishable neural mechanism, a distinct prerequisite chain, and a separable assessment approach. It is not derived from, nor validated by, any psychometric or factor-analytic procedure. Researchers treating this taxonomy as a measurement framework should develop and validate instruments before drawing structural inferences from the domain or skill count.

3. **Domain boundaries are functional hypotheses.** The nine domains represent proposed clusters of mechanistically related skills. The domain-specificity finding for metacognition (Morales et al., 2018; Fleming et al., 2014) provides partial neurobiological support for treating monitoring domains as distinct, but does not validate the specific domain divisions used here. The Self-Boundary Integrity domain (B2.1–B2.4) in particular combines skills

that may prove to be dissociable at the neural level, and whose co-clustering is currently motivated by theoretical coherence rather than empirical convergence.

3. Neuroscientific Foundations

3.1 Neuroplasticity

Neuroplasticity is a family of related processes operating at different scales and timescales. Synaptic plasticity following Hebb's rule (Hebb, 1949) provides the foundational mechanism: temporally correlated pre- and post-synaptic activity produces long-term potentiation (LTP), strengthening synaptic connections; uncorrelated or suppressed activity produces long-term depression (LTD), weakening them. Both are necessary: learning requires simultaneous strengthening of relevant and suppression of irrelevant patterns.

At the structural level, cortical thickness changes are a measurable correlate of sustained training. Right insula and somatosensory cortex thickness increases with concurrent alexithymia reductions were demonstrated after 8-week mindfulness-based stress reduction (MBSR) (Calderone et al., 2024). Hippocampal volume increases from the same intervention likely reflect neurogenesis in dentate gyrus, a process enhanced by exercise and reduced by chronic glucocorticoid exposure. Functional plasticity — changes in how existing circuits are activated without gross structural change — can occur within sessions and is the mechanism underlying network reconfiguration efficiency improvements demonstrated by Yue et al. (2023).

A critical constraint on all of the above is the behavioural relevance gate. Plasticity does not occur through repetition alone. The cochlear implant literature (Fallon et al., 2009) established that cortical reorganisation requires inputs that are behaviourally meaningful to the organism: "Not just any input into the wetware will result in appropriate changes: the input must contain behaviourally important information." Visual prosthetics research confirms the same principle, adding that task variability, appropriate difficulty level, and graded feedback all modulate plasticity durability, with positive effects persisting up to two years when attentional neuromodulators are engaged (Beyeler et al., 2017). This constraint is operationalised in the framework as I1.4 (Practice Quality Monitoring): without it, Layer 3 training can be high-volume and neuroplastically inert.

Individual differences in plasticity capacity are substantial and poorly characterised. Cochlear implant outcome variance is approximately 80% attributable to individual "wetware" differences not accounted for by known factors (Fallon et al., 2009). The framework accommodates this by using individual-level threshold demonstration rather than population-level statistics and by requiring Phase 0 assessment to calibrate entry.

3.2 Attention Regulation Networks

Three large-scale brain networks govern attentional self-regulation. The *Default Mode Network* (DMN — medial PFC, posterior cingulate cortex (PCC), angular gyrus, hippocampus) is active during self-referential thought, autobiographical memory, future simulation, narrative construction, and mind-wandering, which occupies approximately 47% of waking time in typical adults (Killingsworth and Gilbert, 2010). The PCC is the DMN's central hub; high PCC activity characterises the mind-wandering and narrative self-generation state.

The *Executive Control Network* (ECN — dlPFC, posterior parietal cortex) is active during goal-directed attention, working memory, and deliberate reasoning. The ECN and DMN are typically anti-correlated at rest: activating one tends to suppress the other. This anti-correlation is the neural architecture of voluntary attentional control. Mindfulness training specifically increases PCC–dlPFC coupling (Ganesan et al., 2025), the neuroimaging correlate of decreased narrative self-generation coupled with increased executive control during focused practice.

The *Salience Network* (SN — anterior insula, anterior cingulate cortex) is the brain's relevance detector and switching system. It detects salient stimuli (internal or external), switches between DMN and ECN modes, and processes interoceptive signals from the body. The SN is the neural substrate of A2.2 (Dual Presence) and S1.2 (Interoceptive Accuracy). Critically, SN functional reconfiguration efficiency improved *specifically* in mindfulness practitioners versus an active sleep hygiene control group (Yue et al., 2023): less "gear-switching cost" to move between internal monitoring and external task engagement.

ECN functional reconfiguration efficiency also improved specifically in the mindfulness group across all three networks (Yue et al., 2023), with the active control group showing declines. This rest-to-task FC similarity measure is the neural basis for M3.2 (State Shifting) as a trainable capacity with a measurable neural correlate.

3.3 Interoception

Interoception is the perception of the internal state of the body. The anterior insula is the primary cortical processing centre, generating what Calderone et al. describes as a moment-to-moment integrated map of the body's state. Insula cortical thickness increases after 8-week MBSR (Calderone et al., 2024) provide structural evidence that this region is trainable.

The critical qualification established by Ganesan et al. (2025) is that focused-attention breath meditation does *not* reliably develop somatic awareness: the SMS-Body subscale showed no significant group difference in a rigorous fMRI neurofeedback RCT despite strong SMS-Mind (cognitive awareness) improvements. Somatic Intelligence skills (S1.1–S1.4) require dedicated somatic practices — body scan, interoceptive tracking — not breath-focused attention alone. This finding directly shapes the framework's practice design: breath meditation is insufficient

for Domain 1.

The standard heartbeat detection paradigm (Schandry, 1981) is used in the progression model as a Track B measure of S1.2 (Interoceptive Accuracy) not because heartbeat detection is the target skill, but because it is a research-validated proxy for the broader interoceptive accuracy construct that correlates with emotional awareness, body image, and anxiety regulation.

3.4 Emotional Processing and Granularity

The amygdala responds to salience broadly — novel, surprising, or emotionally significant stimuli of any valence — rather than functioning as a single-purpose threat detector. Emotional processing involves competition between bottom-up signals (amygdala, insula, brainstem: fast, automatic) and top-down modulation (PFC, ACC: slower, deliberate). This modulation requires resources: cognitive capacity, attentional bandwidth, low baseline arousal. When PFC resources are depleted by high cognitive load, fatigue, or acute stress, bottom-up processes dominate. This is the neurological basis for I1.1 (Skill-Under-Load) as a cross-layer requirement.

Emotional granularity — the precision with which one deploys emotional concepts — modulates regulation quality (Barrett, 2017). People who can distinguish shame from guilt, dread from apprehension, or tenderness from gratitude show measurably better emotion regulation outcomes because more precise categories produce more precisely targeted regulatory responses. E1.3 (Emotional Vocabulary Expansion) is, at this level, structural skill training, not semantic enrichment.

Amygdala reactivity decreases after 8-week MBSR (Gkintoni et al., 2025), supporting emotional gradient detection (E1.4) at the level of automatic processing prior to explicit labelling. MBCT's established mechanism for depression relapse prevention relies on teaching practitioners to observe negative emotional-cognitive cascades as events rather than facts — a mechanism that operationally requires E1.4 to interrupt the cascade early.

3.5 Default Mode Network and Self-Referential Processing

The DMN constructs the sense of a continuous, coherent self across time through mPFC, PCC, and hippocampal integration. This has a non-trivial implication: the narrative self — one's beliefs about who one is, what one's history means, and what is authentically true for oneself — is a neurological construction rather than a direct perception. As such, it is revisable through targeted experience. This is the biological basis for Layer 3 existing at all.

Mind-wandering correlates with negative affect regardless of the content of the wandering (Killingsworth and Gilbert, 2010). Rumination — DMN activity stuck in perseverative self-referential loops — is the pathological extreme and the mechanism underlying depression,

anxiety, and chronic stress. MBCT's evidence base across 87 studies (Gkintoni et al., 2025) rests on interrupting the automatic coupling between a negative mood and a DMN ruminative cascade. C2.6 (Narrative Monitoring) and P3.2 (Pattern Interruption) are, at the network level, the capacity to detect and break this loop before it completes.

B2.4 (Identity Stability Monitoring) addresses the specific vulnerability that sustained exposure to particular narratives, social environments, or influence campaigns can gradually reshape DMN-mediated identity construction. Because each incremental step is small, slow drift is the hardest variant of external influence to detect. The framework responds by requiring longitudinal self-description documentation as a necessary baseline for B2.4 — a Track A measure that enables detection only when compared against a stable reference.

3.6 Metacognition

Metacognition is cognition about cognition: monitoring, evaluating, and regulating one's own mental processes at a level above the processes themselves. Its neural architecture is broadly prefrontal but not uniform. The rostral anterior cingulate cortex (rACC) and medial PFC monitor cognitive processes and generate confidence and uncertainty signals. The inferior frontal gyrus (BA47) shows sustained and increasing activation in advanced meditators (Calderone et al., 2024), rather than habituating — the neural signature of deepening metacognitive engagement rather than automation.

Introspective accuracy is meaningfully skilled and partially trainable, but has hard limits: some cognitive processes are computationally inaccessible to metacognitive reporting even in highly trained individuals. The framework does not claim to make all processing transparent — it trains the accessible monitoring layer within its genuine limits.

In predictive processing terms (Barrett, 2017), metacognition involves monitoring the prediction-update cycles themselves: noticing when a model is being updated, when a prediction error is being suppressed, or when a prediction is being confused with a percept. C2.5 (Assumption Detection) is the behavioural correlate of this: catching when an inference is being treated as an observation.

3.7 Neuromodulation and the Conditions for Plasticity

Four neuromodulatory systems gate whether neural activity translates into lasting structural change:

Dopamine (DA), released by the VTA in response to unexpected rewards or reward-predicting cues, opens the LTP window at synaptic sites. This is the neurochemical basis for I1.4: engaging, meaningful practice produces more lasting change than boring repetition, and gamification

of learning contexts enhances plasticity durability persisting up to two years by recruiting dopaminergic reinforcement (Beyeler et al., 2017).

Acetylcholine (ACh), released by the basal forebrain during focused attentional engagement, sharpens cortical signal-to-noise ratios and signals "this input matters, encode it." Attention training (Domain 3) is partly training the ACh-modulated gain enhancement system. Without genuine attentional focus, the plasticity window does not open at relevant sites — making Domain 3 a hard biological precondition for effective Layer 2 and Layer 3 training.

Norepinephrine (NE), released by the locus coeruleus in response to arousal signals, follows an inverted-U relationship with cognitive performance (Yerkes-Dodson law): optimal NE levels support sharp, engaged processing; excess NE impairs it. Working at too difficult a performance level impedes learning through NE overshoot (Beyeler et al., 2017), constraining how stress-inoculation training must be graduated.

Brain-Derived Neurotrophic Factor (BDNF), increased by aerobic exercise and mindfulness practice (Calderone et al., 2024) and decreased by chronic glucocorticoids, is required for many forms of LTP and supports hippocampal neurogenesis. Factors like exercise level, sleep quality, and chronic stress are not peripheral to mental training — they modulate the molecular conditions under which plasticity occurs.

3.8 Source Monitoring: The Cognitive Mechanism of Self-Boundary Integrity

Source monitoring is the set of cognitive processes through which people attribute the origins of their memories, beliefs, thoughts, and impulses to a particular source (Johnson et al., 1993; Mitchell and Johnson, 2009). This is the mechanism underlying the framework's Self-Boundary Integrity domain (B2). Every B2 skill is an application of source monitoring applied to different layers of mental content.

Memories and beliefs are reconstructed from stored feature bundles — perceptual detail, spatial and temporal context, semantic content, emotional tone, and records of the cognitive operations involved in generating them. Source attribution is an inference, not a direct read-out. Two modes operate:

Heuristic monitoring (fast): rapid evaluation of memory quality. High perceptual detail reads as "probably perceived"; low contextual specificity reads as "probably self-generated." Fast, low-effort, and error-prone.

Systematic monitoring (slow): deliberate retrieval and plausibility checking against known facts, values, and autobiographical context. Accurate but resource-demanding, and degraded specifically by active working memory manipulation (Maniscalco and Lau, 2015).

The five documented source monitoring failure modes each constitute a specific vulnerability

in the framework's defensive mission:

1. **Source amnesia:** Retrieved content without its source. Feels like prior knowledge or genuine opinion. The default for information encountered without deliberate source-encoding.

2. **Misattributed familiarity:** Repeated exposure increases processing fluency, which heuristic monitoring misreads as evidence of authentic personal belief. This is a primary mechanism of propaganda and large-scale social influence.

3. **Cryptomnesia:** An externally encountered idea experienced as one's own original thought. The external encounter is forgotten; the idea returns feeling internally generated.

4. **Incorporation of fiction as fact:** Narratively processed content is encoded with rich perceptual and emotional detail through narrative simulation. This richness causes heuristic monitoring to subsequently misclassify the content as perceived rather than encountered via story.

5. **Criteria relaxation under load:** Under conditions of high arousal, fatigue, or cognitive demand, systematic monitoring is bypassed. Heuristic monitoring runs alone with lax criteria (Maniscalco and Lau, 2015). This is the mechanism most directly addressed by I1.1 (Skill-Under-Load) training.

Two neurobiologically distinct sub-processes are involved, with different sequencing implications (Simons et al., 2008):

Context monitoring ("was this perceived or did I generate it?") is less medial-PFC dependent and is more accessible to training through basic thought observation practices. It is developed through C2.1 (Thought Observation) and E2.1 (Emotional Source Attribution).

Origin monitoring ("did I generate this, or did another person?") specifically recruits medial anterior PFC (BA10) and the mentalizing network — the circuits that represent other people's mental states (Simons et al., 2008; Lagioia et al., 2011). This is the harder, more PFC-dependent operation and the precise mechanism of B2.1 (Self-Other Signal Distinction).

Individual differences in origin monitoring capacity correlate with psychosis-proneness ($r = -0.56$) across healthy individuals (Simons et al., 2008), establishing this as a continuous trait in the normal population, not a clinical binary. Approximately 13% of the healthy population has bilateral paracingulate sulcus (PCS) absence (Buda et al., 2011), associated with constitutively lower reality monitoring accuracy *and* reduced metacognitive awareness of that deficit. For these individuals, Track B performance assessment is the only reliable measurement channel.

Critically, this circuit is trainable: 80 hours of bottom-up perceptual precision training in schizophrenia patients restored mPFC activation and normalised origin monitoring accuracy,

with mPFC normalisation at training completion predicting social functioning at six-month follow-up (Subramaniam et al., 2012). This provides the strongest direct evidence that the L1 perceptual and attentional substrate enables higher-order boundary monitoring through transfer — confirming the L1→L2 architecture of the framework.

Additional design constraints from source monitoring research: Moseley et al. (2018) demonstrated that internally generated content encoded with high vividness was subsequently more likely to be misattributed as externally perceived. Any framework practice using mental imagery must include an explicit source-marking step during encoding to prevent the vividness-misattribution coupling from strengthening. Bang and Fleming (2018) established that the anterior PFC integrates confidence signals with non-perceptual contaminants — confidence history, current arousal, social context — that degrade source attribution accuracy under influence conditions.

3.9 Two-Tier Metacognitive Architecture and Domain-Specificity

Three converging studies establish that metacognition is not a unified capacity (Morales et al., 2018; Shekhar and Rahnev, 2018; Fleming et al., 2014):

Level 1 (domain-general): Encoded in the dACC/insula/cingulo-opercular network. Generates a generic confidence and accuracy signal that partially predicts performance across all cognitive domains. A classifier trained on one domain's patterns predicts confidence in a different domain. This layer is the neural substrate of I1.4 (Practice Quality Monitoring) and partially transfers across domains.

Level 2 (domain-specific): Encoded in the aPFC/right rostrolateral PFC (BA10). Carries domain-specific metacognitive accuracy representations that do *not* generalise. Individual metacognitive scores for perception and memory are uncorrelated ($r = -0.076$) (Morales et al., 2018). Anterior PFC lesions selectively impair perceptual metacognition while leaving memory metacognition intact — causal, not merely correlational, evidence for domain-specificity (Fleming et al., 2014).

A large-sample fMRI study ($n = 308$) found that higher metacognitive accuracy correlates with *less* aPFC activation (Molenberghs et al., 2016): skilled metacognition is neurally efficient, not effortful. Early monitoring is effortful; competence feels like the same accuracy with less cost. The same study found that acknowledged uncertainty activates OFC/insula (associated with negative affect), while false confidence activates bilateral striatum (dopaminergic reward circuits). This creates a motivated bias toward false certainty that calibration training must address through explicit tolerance of acknowledged not-knowing — an emotional regulation requirement embedded within metacognitive skill development, and a reason why E1 (Emotion Granularity) is a prerequisite for reliable Layer 2 self-monitoring.

The critical practical implication: somatic monitoring skill (S1 domain) does not substitute for emotional monitoring skill (E2 domain), which does not substitute for cognitive monitoring (C2 domain), which does not substitute for self-boundary monitoring (B2 domain). Each requires its own dedicated training and independent assessment.

Landmark Convergent Findings

Table **??** summarises the eight most consequential convergent findings for framework design. Each finding simultaneously constrains what the framework claims and motivates a specific structural choice. The full cross-paper convergence table appears in Section 11.

Table 2: Landmark Convergent Findings and Their Structural Implications (condensed; full detail in Table 4)

Finding	**Key Sources**	**Structural Implication**
Insula/SN is the trainable interoceptive substrate; breath meditation insufficient for somatic awareness	Calderone et al. (2024); Yue et al. (2023)	Dedicated somatic practices required in Layer 1
ECN–DMN dynamics are the substrate of attentional self-regulation	Ganesan et al. (2025); Yue et al. (2023)	Domain 3 attention training is a biological precondition for Layer 2
PFC-dependent regulation fails under acute physiological stress	Raio et al. (2013)	Skill assessment under load is required; calm-only evaluation is insufficient
Source monitoring failure modes are the mechanism of influence vulnerability	Johnson et al. (1993); Mitchell and Johnson (2009); Simons et al. (2008)	B2 domain structured around five specific, named failure modes
Metacognitive accuracy is domain-specific; does not transfer across domains	Morales et al. (2018); Fleming et al. (2014)	Each monitoring domain requires independent training and assessment
mPFC (BA10) origin monitoring circuit is trainable via bottom-up perceptual training	Subramaniam et al. (2012)	Layer 1 perceptual substrate enables Layer 2 boundary monitoring
Practice ratio predicts resilience outcomes; minimum effective dose $\approx$11 min/day	Jha et al. (2015, 2016, 2017, 2020)	Quality and practice ratio are operationally critical variables
Neural and behavioural change are decoupled within the same measurement window	Yue et al. (2023)	Two independent measurement tracks are required throughout

4. The Three-Layer Sequence: Neurological Justification

4.1 Layer 1 to Layer 2 Dependency

Layer 1 builds the sensors: interoceptive accuracy (insula, salience network), emotional signal quality (amygdala-to-insula mapping, emotional granularity), and attentional stability (ECN–DMN anti-correlation). Layer 2 builds the monitoring system: prefrontal metacognition observing the outputs of Layer 1 processes.

The key dependency is that if the sensors are noisy or absent, metacognition has nothing reliable to observe. A person with significant alexithymia attempting C2.1 (Thought Observation) will be working with heavily filtered, inaccurate input. Their "observation" of thoughts will be partially observation and partially construction, and I1.3 (Calibration Honesty) will be correspondingly degraded.

The Phase 0 screening for alexithymia (TAS-20) and interoceptive accuracy (Heartbeat Detection Task (Schandry, 1981)) implements this dependency: before training the monitoring system, verify that the signals it will monitor are sufficiently clear.

Acetylcholine provides a second dependency: Domain 3 attention training is partly training the ACh-modulated gain enhancement system. Without sustained, directed attention on internal processes, metacognitive signals are too noisy to be reliably read. Domain 3 is not a soft prerequisite — it is a hard biological precondition.

4.2 Layer 2 to Layer 3 Dependency

Layer 3 involves deliberate modification of cognitive and emotional patterns. At the neural level, this means intentionally targeting circuits for LTP or LTD through directed, behaviourally relevant practice. The critical point: you cannot reliably modify a circuit you cannot accurately perceive. If Layer 2 monitoring is unreliable, Layer 3 practice will produce change, but the practitioner will be unable to verify whether the target pattern is actually changing, whether an adjacent unintended pattern is also being modified, or whether the modification is producing a harmful side effect.

The history of psychological intervention contains numerous examples of techniques that produce changes — but not the changes intended, or with unmonitored side effects. The difference between systematic skill training and potentially harmful self-modification is whether the monitoring system (Layer 2) is reliably operational before modification work begins. This is the mechanism behind the framework's safety warning at Layer 3: "Operating here without solid Layer 2 capability means modifying patterns you cannot accurately perceive. This is the layer where harm becomes possible."

4.3 The L2-to-L3 Gate as a Safety Requirement

The framework's L2→L3 gate — requiring demonstration of skill under load conditions including mild physiological arousal, a PHQ-9 score below the clinical moderate threshold, and absence of active crisis indicators — is not a curriculum checkpoint. It is a functional safety requirement grounded in: (1) the stress-transfer problem (Raio et al., 2013) requiring that skills be tested under the conditions in which they must operate; (2) the known PFC vulnerability to glucocorticoid-mediated impairment during depressive episodes (Gkintoni et al., 2025); and (3) the pattern modification risk identified above. The gate cannot be waived without removing these safety functions.

5. Probabilistic Monitoring Architecture: Channels, Mechanisms, and Known Failure Modes

5.1 Channels of Subconscious Influence

Any technique capable of subconscious influence must operate through a limited set of channels: (1) bottom-up sensory input generating an emotional response before cortical awareness; (2) source monitoring failure — externally introduced content misattributed as self-generated; (3) narrative priming that makes certain thoughts feel natural, true, or self-generated; (4) emotional contagion shifting affective state before attribution occurs; (5) arousal manipulation that degrades PFC function; and (6) slow identity drift through sustained exposure to identity-incompatible content.

Emotional contagion warrants specific attention as a framework target. It operates through multiple simultaneous mechanisms: facial mimicry activating corresponding muscle groups and generating associated interoceptive signals; limbic resonance evidenced in inter-brain theta and beta coherence during social interaction (Chen et al., 2022); and parasocial/media-mediated contagion exploiting narrative simulation mechanisms. E2.2 (Emotional Contagion Awareness) is the first-line detection skill for this channel, requiring the prerequisite E1.4 (Emotional Gradient Detection) to catch the early signal.

5.2 Detection and Resistance by Channel

Influence Channel	Detection Mechanism	Resistance Mechanism
Bottom-up emotional signal	E1.4 gradient detection + E2.1 source attribution	B2.1 self-other distinction; M3.4 response modulation before full capture

Influence Channel	Detection Mechanism	Resistance Mechanism
Source monitoring failure	B2.1 origin monitoring + C2.5 assumption detection + B2.2 incongruence vs. values	Systematic source-checking habit; values document as external anchor (B2.3)
Narrative priming	C2.6 narrative monitoring + C2.5 assumption detection	B2.2 incongruence detection; P3.2 pattern interruption breaks the chain
Emotional contagion	E2.2 contagion awareness + S1.3 baseline deviation	A2.2 dual presence during exposure; B2.3 value anchoring
Arousal manipulation	S1.4 arousal detection + C2.4 cognitive load sensing	M3.1 voluntary relaxation (parasympathetic, PFC-independent); I1.2 recovery speed
Slow identity drift	B2.4 identity stability monitoring + I1.3 calibration honesty	Longitudinal baseline documentation; topology map as reference structure

Table 3: Detection and Resistance Mechanisms by Influence Channel

5.3 The Stress-Transfer Problem

Raio et al. (2013) demonstrated experimentally that cognitive emotion regulation strategies — reappraisal and acceptance, both taught and practised under calm conditions — produced *zero* reduction in conditioned fear responses when acute physiological stress was induced before application. The stressed group's fear responses were statistically equivalent to pre-training acquisition levels despite having received identical training to the non-stressed group.

The mechanism is noradrenergic, not motivational or effort-related. Acute sympathetic arousal rapidly degrades dorsolateral PFC function through NE overshoot. The same PFC systems that support deliberate reappraisal, perspective-taking, pattern interruption, and all Layer 2 and Layer 3 skills that rely on cognitive override are impaired. This is a state-dependent neurobiological constraint, not a training intensity problem.

The practical implications are severe: Layer 2 and Layer 3 skills assessed only under calm conditions are measuring a capability that is *less relevant* than real-world deployment capability. The conditions under which the framework's monitoring system must operate — arousal, social pressure, emotional activation — are precisely the conditions under which PFC-dependent

strategies are most impaired.

Mitigations (not eliminations):

- Stress-inoculation training is required in Phases 7–8: skills must be practised immediately after mild physiological arousal induction (vigorous exercise, cold exposure, mild social challenge), not only from calm baseline.
- The L2→L3 gate specifically requires demonstration of skill under physiological arousal condition (Condition D in the B8 Load-Condition Assessment), not only under cognitive or emotional load at rest.
- Somatic and parasympathetic regulation (M3.1 — operating downstream of PFC) are less vulnerable than cognitive override approaches. Layer 1 somatic grounding is the fastest available stress-response that does not depend on PFC resources.
- Skills trained toward automaticity through sustained Layer 3 practice reduce PFC cost per deployment, making them more stress-resistant. But automaticity must be *tested* under arousal, not assumed from calm-condition performance.

5.4 Additional Structural Limits

Extreme arousal (acute fear, rage) bypasses PFC entirely — no trained skill survives amygdala hijack intact. The framework mitigates this through proactive arousal detection and regulation (M3.3) before conditions deteriorate, not recovery during them.

Metacognitive monitoring is suppressed during active working memory manipulation (Maniscalco and Lau, 2015). Ruminative, planning, and worry states involve exactly this kind of manipulation: the states in which practitioners most want to self-monitor are neurobiologically the states in which self-monitoring is least reliable. Layer 1 somatic grounding and attentional stabilisation must precede metacognitive attempts when the individual is in an active-manipulation cognitive state. This is documented as a named failure mode in the framework's progression model, not left as an implicit limitation.

Sufficiently subtle influence may not produce detectable deviation: if influence is calibrated to stay within the practitioner's noise threshold, B2.1 will not fire. Detection is probabilistic, not absolute.

Trained monitoring skills require ongoing maintenance. Skills that are not practised decay, with structural changes (cortical thickness, connectivity) persisting longer than functional habits. This makes Phase 8 (ongoing 12-week checkpoints) a design necessity, not a rounding off of the programme.

6. Critical Design Constraints

6.1 Domain-Specificity of Metacognition

The domain-specific layer of metacognition (aPFC, BA10) carries domain-specific representations that do not generalise (Morales et al., 2018; Fleming et al., 2014). A practitioner with excellent Cognitive Process Monitoring (C2 domain) cannot be assumed to have equivalent Emotional Self-Monitoring (E2), Self-Boundary Integrity (B2), or Somatic monitoring (S1). Each domain requires its own dedicated training and must be independently assessed. This directly constrains the framework's assessment design: gate criteria in each domain are assessed independently, and discrepancies between domains are surfaced explicitly rather than averaged away.

6.2 Metacognitive Unavailability Under Working Memory Load

Active working memory manipulation — as distinct from mere maintenance — specifically degrades metacognitive sensitivity (Maniscalco and Lau, 2015). Ruminative, planning, and worry states involve this manipulation. The framework's response is: (1) name this explicitly as a failure mode in practitioner training, so that practitioners know when they are attempting monitoring in a state that prevents reliable monitoring; (2) design Layer 1 grounding practices as the initial response to this state, restoring preconditions before Layer 2 monitoring is attempted.

6.3 Behavioural Relevance as a Plasticity Gate

Plasticity requires behaviourally meaningful input (Fallon et al., 2009; Beyeler et al., 2017). "Not just any input into the wetware will result in appropriate changes." Quantity of practice without quality does not produce structural change. The framework implements I1.4 (Practice Quality Monitoring) as a required cross-layer skill introduced at the Layer 3 entry point, with specific quality conditions: behavioural relevance, appropriate challenge level (not too easy, not overwhelming), contextual variety, and intentional engagement. Practitioners without I1.4 can generate high-volume, neuroplastically inert sessions.

Four independent studies by Jha et al. (2015), Jha et al. (2016), Jha et al. (2017), and Jha et al. (2020) converge on identifying a minimum effective dose of approximately 11 minutes per day of formal practice and, critically, that practice *ratio* (proportion of time practised vs. time given) predicts cognitive resilience outcomes more strongly than total dose. Below threshold, decline is the default; the framework trains against decline, not only for enhancement.

6.4 The BCT Inversion

The Breath Counting Task (BCT) accuracy declined in the group showing the strongest neural and subjective improvement in Ganesan et al. (2025). The likely mechanism: the trained group developed genuine decentering — awareness of the act of counting as a mental process — which interfered with the mechanical execution of counting. This is not attentional failure; it is a shift from automatic execution to meta-aware execution. The BCT and related standard neuropsychological attention tests should not be used as primary skill progress markers; multiple reviews report inconsistent improvement after 8-week MBIs (Calderone et al., 2024; Gkintoni et al., 2025). Neural efficiency measures (ECN reconfiguration, PCC–dlPFC coupling) or ecologically valid tasks are preferred.

7. Training Design Principles

The framework's 35 named practices are organised across a Phase 0–8 progression with weekly schedules and phase exit assessment checklists. Core design principles derived from the evidence review:

1. **Dedicated somatic practices are required.** Breath-focused meditation is insufficient for Domain 1 skills (Ganesan et al., 2025). Body scan and interoceptive tracking must be explicitly scheduled, not assumed to be covered by seated meditation.

2. **Load-condition training is required.** Skills trained only in calm, low-distraction conditions will not transfer to the conditions in which they are needed (Raio et al., 2013; Beyeler et al., 2017). Each domain's skills must be practised under conditions that approximate deployment conditions.

3. **Context monitoring before origin monitoring.** B2.1 (origin monitoring) is trained after C2.1 (thought observation) and E2.1 (emotional source attribution), following the neurobiologically supported sequencing: context monitoring first (less medial-PFC dependent), then origin monitoring (Simons et al., 2008).

4. **Targeted pattern interruption requires targeted training.** Standard mindfulness does not reduce impulsivity in naive practitioners (Calderone et al., 2024; Gkintoni et al., 2025). P3.2 (Pattern Interruption) requires deliberate interruption training aimed at the specific habit chains identified through topology mapping (P3.4), not general mindfulness practice.

5. **Source-marking is required in all visualisation practices.** Any practice using mental imagery must include an explicit "this is self-generated content" step during encoding, to prevent the vividness-misattribution coupling from strengthening (Moseley et al., 2018).

6. **Two independent tracks must be maintained throughout.** Track A (self-report) and Track B (physiological/performance measures) are not substitutes. Brain and behavioural changes are uncorrelated within the same measurement window (Yue et al., 2023), and approximately 13% of the population has a structural deficit that makes Track A systematically biased (Buda et al., 2011). Both tracks are required.

7. **The minimum effective dose is approximately 11 min/day** of formal practice, with the critical variable being practice *ratio*, not total hours (Jha et al., 2015, 2016, 2017, 2020). Quality, not quantity, determines plasticity outcomes (Fallon et al., 2009).

8. Assessment Framework

The assessment framework comprises six validated instruments (Toronto Alexithymia Scale TAS-20; FFMQ Observing Subscale; SMS-Mind; PHQ-9; DASS-21; Rumination Response Scale) and eleven custom protocols (B1–B11, covering somatic baseline, heartbeat detection, arousal calibration, vocabulary audit, attention sustaining, direction and discrimination, distraction resistance, meta-attention probe, dual presence performance, load-condition matrix, and calibration honesty check). Nineteen journal templates standardise longitudinal self-report. Phase gate checklists for all nine transitions formalize threshold requirements.

The B8 Load-Condition Assessment warrants specific notation: skills are rated under five conditions (rested baseline plus four load conditions — cognitive, emotional, social, and physiological arousal). Condition D (physiological arousal, induced immediately after vigorous exercise or cold exposure) is required for the L2→L3 gate, implementing the stress-transfer problem mitigation directly in the assessment design rather than leaving it to practitioner discretion. Retention is scored as mean skill rating across all four load conditions as a percentage of rested baseline.

Criterion types are distinguished in the gate system: *auto* criteria are computed directly from assessment results; *semi-auto* criteria are computationally evaluable but require explicit practitioner acknowledgment (used for crisis indicators); *manual* criteria require practitioner attestation from journal-based, longitudinal, or behavioural evidence. Gate 7 (L2→L3) contains PHQ-9 and crisis-indicator criteria flagged as *hard blocks* (the gate cannot pass if these criteria fail, regardless of other scores).

PHQ-9 gate rationale. The PHQ-9 hard block at score ≥ 10 (moderate severity) warrants explicit justification, as threshold choice and instrument selection involve calibration decisions that may appear policy-driven without explanation.

Why PHQ-9 specifically. The PHQ-9 is selected over alternative symptom measures because it tracks the depressive dimension most directly linked to the PFC functional impairment that

makes Layer 3 work unsafe. Glucocorticoid elevation associated with clinically significant depression produces measurable dlPFC volume reduction and connectivity decrements (Gkintoni et al., 2025), directly compromising the monitoring reliability Layer 3 requires. The PHQ-9 is brief (nine items, <5 minutes), extensively validated, and widely used in self-administered screening contexts. The DASS-21 is already present in the battery for dimensional profiling of depression, anxiety, and stress; PHQ-9 adds a dichotomised severity check calibrated to the specific threshold at which clinical support is typically indicated.

Why ≥ 10 *and not* ≥ 8. PHQ-9 ≥ 10 is the established clinical cut-off for moderate depression severity, where functional impairment becomes clinically significant. The neurobiological rationale targets the PFC impairment level at which monitoring reliability is compromised, not the first appearance of depressive symptoms. Lowering the threshold to 8 would block practitioners in the mild-symptom range who have demonstrated functional Layer 2 competency and for whom continued training under clinical support may be the appropriate path; it would also increase false-positive gate blocks substantially without commensurate safety gain. The ≥ 10 threshold is conservative without being prohibitive, and matches existing clinical guidance on when professional referral is indicated.

Why not GAD-7. Anxiety disorders and generalised anxiety do not show the same pattern of PFC monitoring compromise as depressive episodes. Moderate anxiety can elevate arousal and degrade performance on specific tasks, but the glucocorticoid-mediated dlPFC volumetric and connectivity reductions associated with moderate-to-severe depression are not equivalently demonstrated for anxiety at the same threshold level. The DASS-21 anxiety subscale is already administered and flags high-anxiety presentations for clinical support; a GAD-7 hard block would duplicate this coverage without adding the specific PFC-impairment rationale that motivates the PHQ-9 block. This is a genuine gap in the current safety screen, and reviewers are correct that anxiety severity screening warrants further consideration.

Why not mania or bipolar screening. The framework does not include a mania or hypomania screen, which is a legitimate limitation. Hypomanic states can produce inflated self-assessment of skill (Molenberghs et al., 2016), precisely degrading I1.3 (Calibration Honesty) in ways that make Layer 3 work unsafe for different reasons than depression. The absence of a brief mania screen (e.g., MDQ or HCL-32) reflects a gap in the current gate design rather than a principled exclusion. This is noted as an open item for protocol refinement.

8.1 Operational Definition of Framework Success

A framework that cannot specify what change would count as success is a conceptual integration, not an empirical programme. The following operationalises the dependent variables, hypothesised neural markers, behavioural endpoints, and expected timeframes against which the Mental Mastery framework should be evaluated.

Measurable behavioural endpoints.

- *Source attribution accuracy:* Pre/post change on standardised reality monitoring tasks (e.g., Johnson & Raye paradigm adapted for social content), with effect size $d \geq 0.5$ constituting a meaningful improvement.
- *Monitoring reliability under load:* B8 retention percentage (skill ratings under load conditions as % of rested baseline) $\geq 80\%$ across all four conditions, including physiological arousal (Condition D). Minimum meaningful change: 10 percentage points from baseline.
- *Metacognitive calibration:* Reduction in mean absolute gap between self-rated and evidence-rated skill level (B11 Calibration Honesty protocol) to ≤ 1.0 on a 0–10 scale. Effect size benchmarks from Molenberghs et al. (2016) suggest the domain-general metacognitive layer is measurable with standard psychometric instruments.
- *Interoceptive accuracy:* Heartbeat detection accuracy (Schandry paradigm (Schandry, 1981)) $\geq 70\%$ across three consecutive weekly sessions.
- *Attentional stability under distraction:* ≤ 2 unrecovered attentional drifts per session across ≥ 3 distinct distractor environments.

Hypothesised neural markers. The framework predicts the following neural changes, based on the mechanisms described in Section 3. These are hypotheses, not confirmed outcomes of the framework specifically:

- *Right insula cortical thickness increase* (Domains 1–2): effect detectable at ≥ 8 weeks, magnitude consistent with Calderone et al. (2024) (≈ 0.1–0.3 mm in active vs. control group).
- *ECN and SN reconfiguration efficiency improvement* (Domain 3, A2.2): rest-to-task FC similarity increase, measurable via fMRI, consistent with Yue et al. (2023) effect sizes.
- *Reduced PCC activation during monitoring tasks* (C2.1, C2.6): shift from identified to observing mode, indexed by SMS-Mind subscale improvement and PCC deactivation during thought observation paradigms (Ganesan et al., 2025).
- *mPFC (BA10) activation increase during source attribution tasks* (B2.1): the specific circuit confirmed by Simons et al. (2008); Subramaniam et al. (2012); measurable with fMRI source-attribution paradigms in healthy populations.

Timeframe expectations. Based on the timescale differentiation established in the reviewed literature:

- *Weeks 1–4:* Subjective improvements in monitoring awareness and attentional stability; self-report measures (FFMQ Observing, SMS-Mind) likely to show change. Structural neural changes not expected.
- *Weeks 4–8:* Functional connectivity reconfiguration measurable (Yue et al., 2023); interoceptive accuracy thresholds reachable. L1 phase exit criteria met for practitioners without significant alexithymia at entry.
- *Weeks 8–16:* Structural changes (insula thickness, hippocampal volume) potentially measurable. Layer 2 monitoring skills approaching threshold in practitioners with consistent practice at required dose (Jha et al., 2015).
- *Months 4–12:* Layer 2 stability under load demonstrable; L2→L3 gate criteria reachable. Metacognitive calibration improvements stabilising.
- *Months 12+:* Pattern Work (Layer 3) outcomes; relapse-prevention analogues measurable. Identity stability monitoring requires longitudinal tracking at this timescale.

What effect size counts as meaningful. A successful framework trial should demonstrate: (a) monitoring reliability (B8 retention) improvement of ≥ 10 percentage points over active control at 8 weeks; (b) source attribution accuracy improvement of Cohen's $d \geq 0.4$ relative to waitlist; (c) PHQ-9 and adverse event monitoring showing no increase in psychological distress. Given the 87% publication positive rate in this literature (Coronado-Montoya et al., 2016), pre-registration and active control conditions are required for any claim to be credible.

9. Individual Differences and Plasticity Variation

The approximately 80% of cochlear implant outcome variance unaccounted for by known factors (Fallon et al., 2009) provides a humbling benchmark. The analogous figure for mindfulness training outcomes is not precisely established, but the heterogeneity across 87 RCTs (Gkintoni et al., 2025) suggests it is similarly large. The framework accommodates this by: avoiding fixed timelines (ranges, not schedules); using Phase 0 assessment to calibrate entry; and requiring individual-level threshold demonstration rather than group-average statistics.

Table 4: Individual Difference Variables Relevant to Training Trajectory

Variable	**Effect**	**Framework Implication**
Alexithymia (TAS-20)	Reduces interoceptive signal quality; slows Domain 1–2 development	Phase 0 screens; extended Phase 1–2 tracks
Baseline trait mindfulness (FFMQ)	Affects Layer 1 entry point	Phase 0 determines entry phase
Age	Plasticity rates lower; higher-order cognition more malleable than early sensory	Timescale expectations lengthen; structural changes may take longer
Chronic stress or trauma	Elevated cortisol baseline impairs PFC; S1.3 baselines may be dysregulated	Clinical support may be required before entry
Bilateral PCS absence (≈13%)	Constitutively lower reality monitoring + reduced metacognitive awareness of deficit (Buda et al., 2011)	Track B only for B2 domain; Track A systematically biased
BDNF Val66Met polymorphism	Reduced activity-dependent BDNF secretion; potentially reduced neuroplastic response	Explains residual outcome variance; not screened
Prior meditation practice	Higher entry point; faster Phases 1–3 progression	Phase 0 captures this
Sleep quality	Plasticity consolidation primarily occurs during sleep; poor sleep degrades LTP	Background variable monitored throughout
Aerobic exercise level	Strongly increases BDNF; enhances hippocampal neurogenesis	Faster structural changes expected with regular aerobic exercise

10. Honest Boundaries

10.1 Supported Claims

The following claims have strong or moderate convergent support:

- Attention regulation is trainable with measurable neural correlates (Yue et al., 2023; Ganesan et al., 2025) — *well supported*
- Interoceptive accuracy is trainable with structural neural correlates (Calderone et al., 2024) — *well supported*
- Emotion granularity and somatic-emotion mapping are trainable (Calderone et al., 2024; Gkintoni et al., 2025) — *well supported*
- Thought observation and decentering are trainable with clinical-level outcomes (Gkintoni et al., 2025; Ganesan et al., 2025) — *well supported*
- Directed neuroplastic practice produces structural brain change given appropriate quality conditions (Fallon et al., 2009; Beyeler et al., 2017; Calderone et al., 2024) — *supported with conditions*
- Source monitoring is the cognitive mechanism of self-boundary integrity (Johnson et al., 1993; Mitchell and Johnson, 2009; Simons et al., 2008) — *well supported at component level*
- The three-layer sequence has neurological justification at the mechanism level — *supported by mechanism; not tested as a complete integrated framework*

10.2 Plausible But Not Strongly Confirmed

- Self-Boundary Integrity skills (B2.1–B2.4) as a coherent trainable domain in healthy populations (individual components confirmed; integrated-system evidence from clinical populations only (Subramaniam et al., 2012))
- Dual Presence (A2.2) sustaining under advanced Layer 3 conditions (theoretical; limited direct testing)
- Identity Stability Monitoring (B2.4) as a mechanism for reduced vulnerability to slow drift (logically compelling; no direct empirical test of the specific skill in healthy populations)
- Transfer from domain-specific metacognitive training to naturalistic influence resistance (source monitoring circuit is trainable (Subramaniam et al., 2012); transfer is inferred, not directly demonstrated)

Direct evidence gap — influence resistance in healthy populations. No study to date has directly tested whether source monitoring training in healthy individuals improves resistance to real-world persuasive or influence contexts. The existing evidence chain runs: (a) source monitoring failure modes are documented mechanisms through which influence operates (Johnson et al., 1993; Mitchell and Johnson, 2009); (b) the mPFC circuit supporting origin monitoring is trainable in clinical populations (Subramaniam et al., 2012); (c) perceptual precision training generalises to higher-order source monitoring in schizophrenia patients (Subramaniam et al., 2012). None of these steps has been demonstrated in healthy adults under naturalistic influence conditions. The framework's claim that training the B2 skill set increases resistance to influence is a mechanistically motivated hypothesis, not an empirical finding. Researchers should not treat the framework's influence-resistance rationale as established fact pending direct experimental evidence.

This gap is not incidental: it is the central empirical question the framework raises and cannot answer with existing data. Studies explicitly measuring B2-domain skill development alongside pre/post exposure to standardised persuasion or social influence paradigms (e.g., illusory truth effect, social proof manipulation, narrative embedding of factual claims) would constitute direct tests of the framework's core hypothesis.

10.3 Claims Exceeding Current Evidence

- The framework's complete effectiveness in reducing vulnerability to advanced neural interface technology (such technology at the implied level does not yet exist; testing is not possible)
- That any training protocol reliably reduces detection failure across all influence channels (documented structural limits in Section 5)
- That measurable Layer 2 competency *guarantees* safe Layer 3 work without residual risk (reduces risk substantially; does not eliminate it)
- That domain-specific metacognitive training transfers across domains — causal evidence from lesion studies shows it does not for the precision monitoring layer (Fleming et al., 2014)

10.4 Publication Bias

87% of mindfulness-based therapy RCTs report positive results (Coronado-Montoya et al., 2016), implying systematic inflation of effect size estimates throughout this literature base. The framework responds by: (1) using conservative threshold specifications in the progression model; (2) treating effect size estimates as upper bounds; (3) requiring both Track A and

Track B evidence (self-report alone is the most inflation-vulnerable); and (4) maintaining I1.3 (Calibration Honesty) as an ongoing cross-layer requirement that applies equally to practitioners and to the framework itself.

11. Convergence Summary

12. Conclusion

The Mental Mastery framework proposes that systematic, sequenced training in 41 specific cognitive skills — grounded in identifiable neural mechanisms and assessed through independent physiological and behavioural measures — can produce a meaningfully more capable cognitive self-governance system than untrained individuals possess. The evidence review across 33 papers supports the core mechanisms, validates the layer sequence as neurologically non-arbitrary, and substantially constrains what training can and cannot accomplish.

The most important novel contribution of the evidence review is the grounding of the Self-Boundary Integrity domain in source monitoring theory (Johnson et al., 1993; Mitchell and Johnson, 2009; Simons et al., 2008): the five failure modes of source monitoring precisely identify the vulnerabilities the framework trains against, and the neural substrate of origin monitoring (mPFC, BA10) is independently confirmed as trainable through bottom-up perceptual and attentional training (Subramaniam et al., 2012). This provides a mechanistic account of the probabilistic monitoring architecture that moves it from a useful metaphor to a testable hypothesis with specified failure modes.

The most important constraint is the stress-transfer problem (Raio et al., 2013): skills assessed only under calm conditions demonstrate substantially lower monitoring reliability under the conditions in which they must operate. This is not a theoretical concern — it has been directly and experimentally confirmed. The framework's mitigations (stress-inoculation training, arousal-condition gate assessment, emphasis on parasympathetic over cognitive-override approaches) are necessary responses, not optional additions.

What remains to be established is whether the specific skill sequences described, practised through the specific protocols defined, and assessed through the specific instruments selected produce the claimed effects in healthy populations at scale. This is an empirical programme, and this preprint is its beginning, not its completion.

Request for Expert Review

This preprint is circulated to solicit expert review of five specific areas:

Table 5: Key Convergent Findings Across 33 Papers and Framework Implications

Finding	Key Sources	Framework Implication
Original 7 Papers		
DMN-ECN dynamics are the neural substrate of attentional self-regulation	Ganesan et al. (2025); Yue et al. (2023)	Domain 3 and 7 grounding confirmed
Insula/SN is the interoceptive and somatic-emotion mapping substrate	Calderone et al. (2024); Yue et al. (2023)	Dedicated somatic practice required
Behavioural relevance gates cortical plasticity	Fallon et al. (2009); Beyeler et al. (2017)	I1.4 is a required cross-layer skill
Brain and behavioural change are decoupled within the same window	Yue et al. (2023)	Two independent measurement tracks required
Thought observation/decentering is the best-evidenced Layer 2 skill	Ganesan et al. (2025); Gkintoni et al. (2025)	C2.1 and C2.6 anchor Layer 2
State shifting is measurable via FC reconfiguration efficiency	Yue et al. (2023)	M3.2 confirmed as trainable
Publication bias is substantial in this literature	Gkintoni et al. (2025)	Effect sizes treated as upper bounds
Key Expansion Findings		
Source monitoring is the cognitive mechanism of the probabilistic monitoring architecture	Johnson et al. (1993); Mitchell and Johnson (2009); Simons et al. (2008)	B2 domain grounded; five failure modes named
mPFC (BA10) is the trainable neural substrate of origin monitoring	Simons et al. (2008); Subramaniam et al. (2012); Buda et al. (2011)	B2.1 has a specific, trainable circuit
PFC-dependent regulation fails under acute physiological stress	Raio et al. (2013)	Stress-inoculation required; L2→L3 gate includes arousal condition
Metacognitive accuracy does not transfer across domains	Morales et al. (2018); Fleming et al. (2014)	Each monitoring domain requires independent training
Practice quantity and ratio, not instruction, drive resilience	Jha et al. (2015, 2016, 2017, 2020)	11 min/day minimum dose; practice ratio is key variable
Metacognitive monitoring suppressed by active WM manipulation	Maniscalco and Lau (2015)	Named failure mode; somatic grounding precedes monitoring attempts
Overconfidence is neurobiologically motivated (striatum reward)	Molenberghs et al. (2016)	Calibration training must include uncertainty-tolerance component
Bottom-up precision training transfers to higher-order source monitoring	Subramaniam et al. (2012)	Strongest direct L1→L2 architecture confirmation
Cognitive decline is the default under sustained high demand; training reduces vulnerability to decline	Jha et al. (2017, 2020)	Early-phase outcome framing: reduced vulnerability to decline, not performance enhancement

1. **Source monitoring grounding (Section 3.8 and Domain B2).** The mapping of source monitoring failure modes to specific B2 skills is based on the reviewed literature but has not been verified by source monitoring specialists. Expert review is requested on: whether the five failure modes accurately represent the current source monitoring literature; whether the sequencing of context-before-origin monitoring is neurobiologically justified; and whether the B2 assessment protocols (B9 Boundary Challenge Assessment, B8 Condition D) validly measure the described constructs.

2. **The stress-transfer problem mitigation (Section 5.3).** The mitigations proposed for the Raio et al. (2013) constraint — stress-inoculation training, arousal-condition gate requirements, emphasis on Layer 1 somatic regulation as less PFC-dependent — are derived from mechanism rather than direct experimental testing of these specific mitigations. Expert review is requested on: whether these mitigations are mechanistically sufficient; whether alternative or additional mitigations are indicated; and whether the evidence from sensory prosthetics research (Beyeler et al., 2017) is appropriately applied to this domain.

3. **Domain-specificity implications for training design (Section 3.9 and 6.1).** The conclusion that each monitoring domain requires independent training and assessment is drawn from domain-specific metacognition research (Morales et al., 2018; Fleming et al., 2014). Expert review is requested on: whether the domain divisions in the framework (somatic, emotional, cognitive, self-boundary) map appropriately onto the neurobiologically-established domain-specificity findings; and whether the degree of independence claimed is warranted.

4. **Assessment validity and clinical thresholds (Section 8).** The B-series custom protocols (B1–B11) are framework-specific instruments without independent validation. Expert review is requested on: whether the threshold specifications for each protocol are defensible; whether the PHQ-9 $\geq$ 10 hard-block is calibrated appropriately; and whether the Track B measures (particularly B8 Load-Condition Assessment and B2 Heartbeat Detection) adequately capture the constructs they are intended to measure.

5. **Layer 3 safety provisions (Section 4.3).** The framework's claim that Layer 2 competency is a necessary precondition for safe Layer 3 work rests on mechanism arguments rather than empirical study of harm rates in practitioners who attempt Layer 3 work without adequate Layer 2 foundation. Expert review is requested on: whether the safety case is adequately specified; whether additional contraindications are indicated; and whether the L2→L3 gate criteria as described are sufficient safeguards or require tightening.

Responses and critiques are welcomed and will be incorporated into the next version. Correspondence should be directed to: *[contact withheld for blind review]*.

References

Bang, J. W. and Fleming, S. M. (2018). Distinct encoding of decision confidence in human medial prefrontal cortex. *Proceedings of the National Academy of Sciences*.

Barrett, L. F. (2017). The theory of constructed emotion: an active inference account of interoception and categorization. *Social Cognitive and Affective Neuroscience*, 12(1):1–23.

Beyeler, M., Rokem, A., Boynton, G. M., and Fine, I. (2017). Learning to see again: biological constraints on cortical plasticity and the implications for sight restoration technologies. *Journal of Neural Engineering*, 14:051003.

Buda, M., Fornito, A., Bergström, Z. M., and Simons, J. S. (2011). A specific brain structural basis for individual differences in reality monitoring. *Journal of Neuroscience*, 31(40):14308–14313.

Calderone, A. et al. (2024). Neurobiological changes of mindfulness-based stress reduction: A systematic review. *Biomedicines*, 12:2613.

Chen, J. et al. (2022). Inter-brain synchrony and social emotion regulation. *Frontiers in Neuroscience*.

Coronado-Montoya, S., Levis, A. W., Kwakkenbos, L., Steele, R. J., Turner, E. H., and Thombs, B. D. (2016). Reporting of positive results in randomized controlled trials of mindfulness-based mental health interventions. *PLOS ONE*, 11(4):e0153220.

Fallon, J. B., Irvine, D. R. F., and Shepherd, R. K. (2009). Cochlear implants and brain plasticity. *Journal of Neural Engineering*, 6:065008.

Fleming, S. M., Ryu, J., Golfinos, J. G., and Blackmon, K. E. (2014). Domain-specific impairment in metacognitive accuracy following anterior prefrontal lesions. *Brain*, 137(10):2811–2822.

Ganesan, K. et al. (2025). Real-time neurofeedback-guided mindfulness training modulates posterior cingulate cortex activity and enhances mindfulness. *Mindfulness*, 16:2787–2807.

Gkintoni, E. et al. (2025). Neurocognitive features of mindfulness-based interventions in mental health: A systematic review. *Journal of Clinical Medicine*, 14:1703.

Hebb, D. O. (1949). *The Organization of Behavior*. Wiley, New York.

Jha, A. P., Morrison, A. B., Dainer-Best, J., Parker, S. C., Rostrup, N., and Stanley, E. A. (2015). Minds “at attention”: Mindfulness training curbs attentional lapses in military cohorts. *PLOS ONE*, 10(2):e0116889.

Jha, A. P., Morrison, A. B., Parker, S. C., and Stanley, E. A. (2016). Examining the protective effects of mindfulness training on working memory capacity and affective experience. *Emotion*.

Jha, A. P., Zanesco, A. P., Denkova, E., Morrison, A. B., Ramos, N., Chichester, K., and Stanley, E. A. (2017). Bolstering cognitive resilience via train-the-trainer delivery of mindfulness training in applied high-demand settings. *Mindfulness*.

Jha, A. P., Zanesco, A. P., Denkova, E., Rooks, J., Morrison, A. B., and Stanley, E. A. (2020). Comparing mindfulness and positivity trainings in high-demand cohorts. *Cognitive, Affective, & Behavioral Neuroscience*, 20:99–115.

Johnson, M. K., Hashtroudi, S., and Lindsay, D. S. (1993). Source monitoring. *Psychological Bulletin*, 114(1):3–28.

Killingsworth, M. A. and Gilbert, D. T. (2010). A wandering mind is an unhappy mind. *Science*, 330(6006):932.

Lagioia, A. et al. (2011). Neural correlates of reality monitoring. *Neuroscience*.

Maniscalco, B. and Lau, H. (2015). Manipulation of working memory contents selectively impairs metacognitive sensitivity in a concurrent visual discrimination task. *Neuroscience of Consciousness*, 2015(1):niv002.

Mitchell, K. J. and Johnson, M. K. (2009). Source monitoring 15 years later: What have we learned from fMRI about the neural mechanisms of source memory? *Psychological Bulletin*, 135(4):638–677.

Molenberghs, P. et al. (2016). Neural correlates of metacognitive ability and of feeling confident: a large-scale fMRI study. *Social Cognitive and Affective Neuroscience*, 11(12):1942–1951.

Morales, J., Lau, H., and Fleming, S. M. (2018). Domain-general and domain-specific patterns of activity supporting metacognition in human prefrontal cortex. *Journal of Neuroscience*, 38(14):3534–3546.

Moseley, P. et al. (2018). Continuums of hallucination-proneness and their relationship with source monitoring and metacognition. *Neuropsychologia*.

Raio, C. M., Orederu, T. A., Palazzolo, L., Shurick, A. A., and Phelps, E. A. (2013). Cognitive emotion regulation fails the stress test. *Proceedings of the National Academy of Sciences*, 110(37):15139–15144.

Schandry, R. (1981). Heart beat perception and emotional experience. *Psychophysiology*, 18(4):483–488.

Shekhar, M. and Rahnev, D. (2018). Distinguishing the roles of dorsolateral and anterior PFC in visual metacognition. *Journal of Neuroscience*.

Simons, J. S., Henson, R. N. A., Gilbert, S. J., and Fletcher, P. C. (2008). Separable forms of reality monitoring supported by anterior prefrontal cortex. *Journal of Cognitive Neuroscience*, 20(3):447–457.

Subramaniam, K., Gill, J., Fisher, M., Mukherjee, P., Turetsky, B. I., and Vinogradov, S. (2012). Computerized cognitive training restores neural activity within the reality monitoring network in schizophrenia. *Neuron*, 73(5):842–853.

Yue, L. et al. (2023). Mindfulness meditation training improves functional connectivity of brain resting-state networks. *Translational Psychiatry*, 13:345.